What Predicted the 2008 Housing Crash?

Bubble Deflation vs. Economic Fundamentals

Introduction

The 2008 financial crisis produced the largest decline in U.S. home prices since the Great Depression, but the damage was not evenly distributed. Nevada lost over 55% of its peak home value; North Dakota lost nothing. Was the crash simply a bubble deflation, or did pre-crisis economic fundamentals determine which states were resilient? We test this using state-level FRED data (house price index, unemployment, income, population) for all 50 states plus DC, with the peak-to-trough HPI decline as the outcome variable.

Beyond the regression analysis, we investigate LLMs as prediction tools for small-sample economic tasks. Claude Opus 4.6 achieves mean RMSE 4.34 across 20 cross-validation splits versus OLS's 8.88. However, a zero-data control (the model receives only state names, no features or training data) achieves RMSE 2.58, indicating the model has memorized crash outcomes from pre-training. This reframes the experiment as a case study in *prior-likelihood interaction*: providing four features that explain 58% of variance degrades predictions from a model whose pre-trained prior is already near-perfect. The result illustrates a general Bayesian principle (incomplete data can hurt when the prior is strong) and identifies the conditions under which the predictive primitive hypothesis could be tested cleanly.

Data wrangling

The raw data consists of 204 CSV files downloaded from FRED, four time series for each of the 50 states plus DC spanning 2000-2012. We read each series, extract summary statistics at key dates, and assemble them into a single cross-sectional DataFrame with one row per state. Full data construction code is available in the project repository.

Computing crash severity from the House Price Index

For each state, we find the peak HPI value in the 2000-2012 window, then find the minimum HPI after that peak date. The crash percentage is `(trough - peak) / peak * 100`. We also

compute pre-crisis appreciation as the percent change in HPI from Q1 2003 to Q1 2006, which measures how much the bubble inflated before it burst.

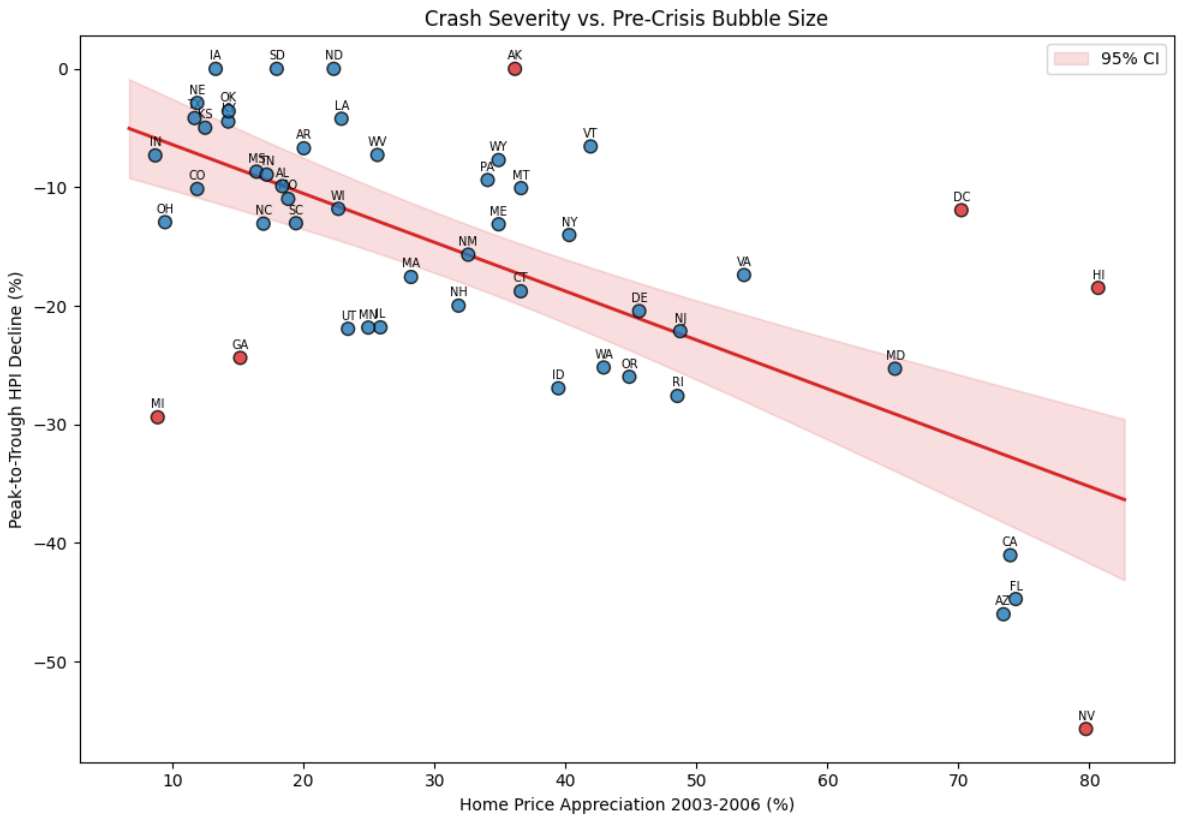
	state	appreciation_03_06	ur_2006	pcpi_2006	pop_growth_03_06	crash_pct
27	NV	79.738252	4.166667	39437	12.175467	-55.694418
2	AZ	73.441020	4.266667	34460	9.414569	-46.013732
8	FL	74.374694	3.283333	38009	6.838974	-44.729060
4	CA	73.961513	4.891667	41454	2.178650	-41.035641
21	MI	8.860281	6.866667	33563	-0.050502	-29.395725
38	RI	48.543015	5.258333	38719	-0.769689	-27.596454
11	ID	39.453794	3.391667	31244	7.722645	-26.951190
36	OR	44.873014	5.391667	34488	3.481644	-25.980296
19	MD	65.156794	3.850000	45293	2.385218	-25.299040
46	WA	42.914120	4.991667	39744	4.368168	-25.194308

Each row is one state. The outcome variable `crash_pct` is the peak-to-trough percentage decline in the FHFA House Price Index (2006-2012). More negative values indicate a more severe crash. The four predictors are pre-crisis indicators observable before the crash began:

- **appreciation_03_06**: Home price appreciation 2003-2006 (%). Proxy for bubble size.
- **ur_2006**: Unemployment rate in 2006 (%). Pre-crash labor market health.
- **pcpi_2006**: Per-capita personal income in 2006 ($). Economic baseline.
- **pop_growth_03_06**: Population growth 2003-2006 (%). Migration-driven demand.

The dataset contains 51 observations (50 states + DC). Crash severity ranges from 0% (no decline: ND, IA) to -55.7% (Nevada).

Visualizations



r = -0.684, R-squared = 0.467

Analysis: does the bubble explain the crash?

```
lm_simple = smf.ols('crash_pct ~ appreciation_03_06', data=df).fit()
print(f'R-squared: {lm_simple.rsquared:.3f}')
print(pd.DataFrame({'coef': lm_simple.params, 'std err': lm_simple.bse,
 ↪  'p-value': lm_simple.pvalues}).round(4))
```

```
R-squared: 0.467
                      coef  std err  p-value
Intercept          -2.2788   2.4212   0.3512
appreciation_03_06 -0.4120   0.0629   0.0000
```

```
lm_multi = smf.ols('crash_pct ~ appreciation_03_06 + ur_2006 + pcpi_2006 +
↪  pop_growth_03_06', data=df).fit()
print(f'R-squared: {lm_multi.rsquared:.3f}')
print(pd.DataFrame({'coef': lm_multi.params, 'std err': lm_multi.bse,
↪  'p-value': lm_multi.pvalues}).round(4))
```

```
R-squared: 0.582
                       coef  std err  p-value
Intercept           10.3390  10.5473   0.3321
appreciation_03_06  -0.3765   0.0751   0.0000
ur_2006             -2.2752   1.2133   0.0671
pcpi_2006            0.0000   0.0002   0.9490
pop_growth_03_06    -1.4420   0.4845   0.0046
```

```
from statsmodels.stats.anova import anova_lm
print(anova_lm(lm_simple, lm_multi))
```

```
   df_resid          ssr  df_diff     ss_diff         F    Pr(>F)
0      49.0  4012.335552      0.0         NaN       NaN       NaN
1      46.0  3151.414654      3.0  860.920898  4.188845  0.010558
```

Model 1 (appreciation only) explains about 47% of the variance in crash severity. Each additional percentage point of pre-crisis appreciation corresponds to roughly a 0.41 percentage point larger crash. Model 2 adds unemployment, income, and population growth, raising R-squared to about 0.58. The F-test above evaluates whether the three additional predictors are jointly significant, that is, whether Model 2 explains significantly more variance than Model 1. If the p-value is below 0.05, the economic fundamentals add predictive power beyond the bubble alone.

Simulation: is the relationship real?

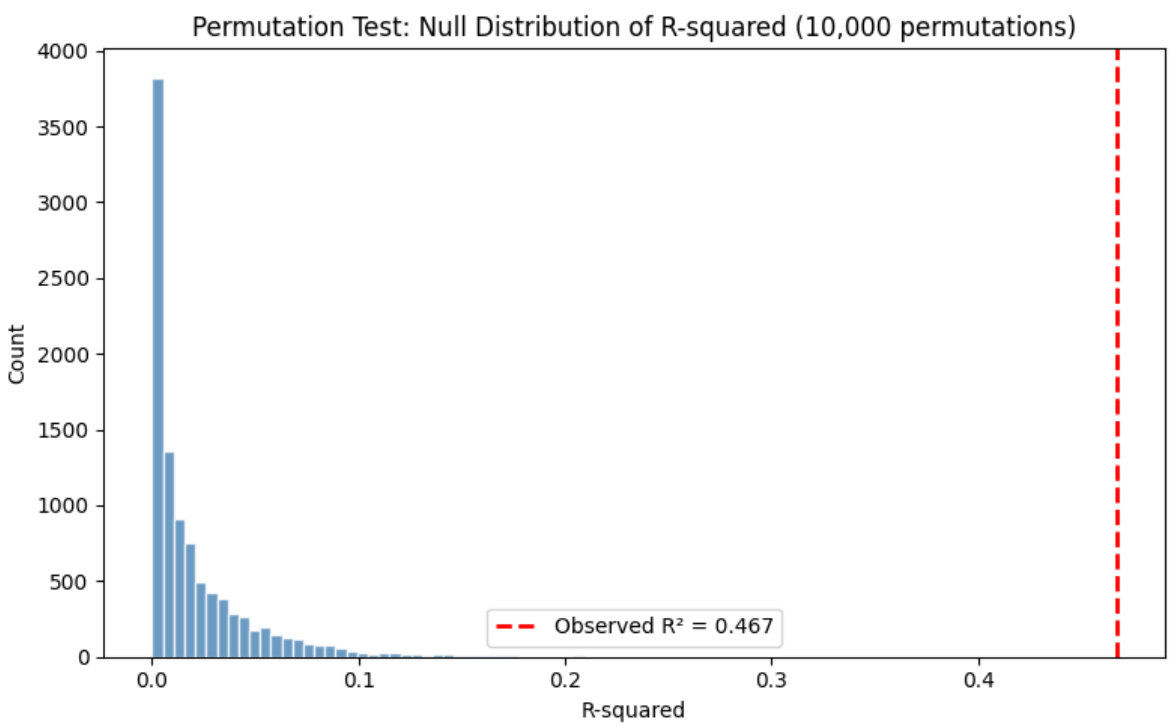
Under the null hypothesis that pre-crisis appreciation has no relationship with crash severity, we would expect any apparent correlation to vanish if we shuffled the crash labels. To test this, we permute `crash_pct` across states 10,000 times, refit the simple regression each time, and collect the null distribution of R-squared values. We then compare our observed R-squared to this distribution.

```
np.random.seed(42)
n_perm = 10_000
null_r2 = np.empty(n_perm)
```

```python
for i in range(n_perm):
    df_perm = df.copy()
    df_perm['crash_pct'] = np.random.permutation(df['crash_pct'].values)
    m = smf.ols('crash_pct ~ appreciation_03_06', data=df_perm).fit()
    null_r2[i] = m.rsquared

observed_r2 = lm_simple.rsquared
p_value = np.mean(null_r2 >= observed_r2)
print(f'Observed R-squared: {observed_r2:.4f}')
print(f'Permutation p-value: {p_value:.4f}')
```

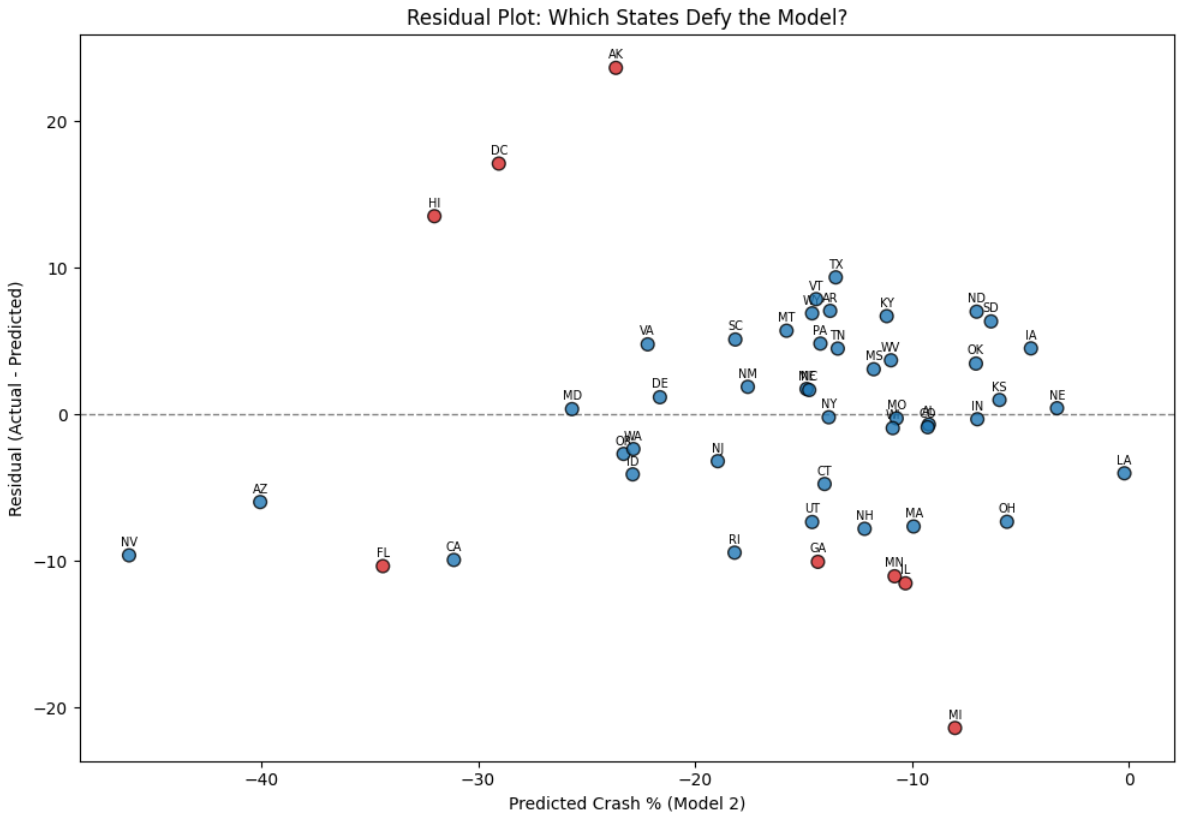
```
Observed R-squared: 0.4672
Permutation p-value: 0.0000
```



The observed R-squared is far above the null distribution, confirming that the relationship between pre-crisis appreciation and crash severity is not an artifact of chance. The bubble-crash link is real.

Which states defy the model?

The regression captures the general pattern, but some states crash much harder or softer than the multiple regression predicts. These outliers show what the model cannot capture: idiosyncratic economic structure that four pre-crisis indicators do not reflect.



Can an LLM agent outperform a regression?

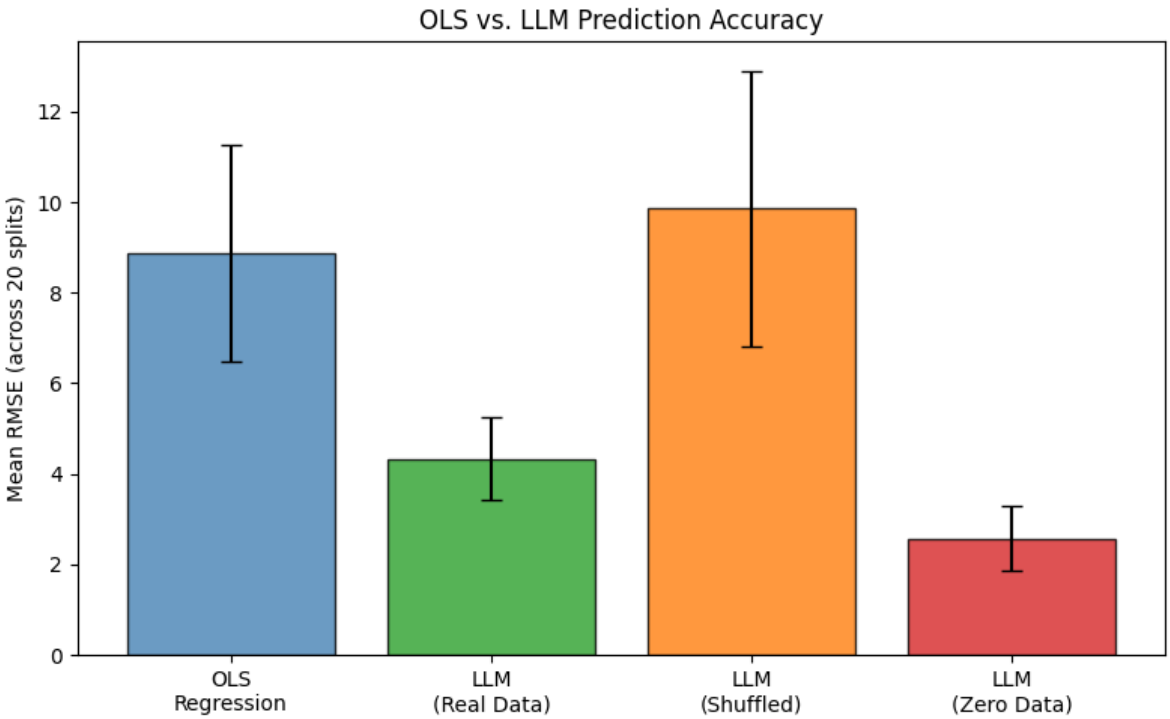
OLS has perfect computation but zero domain knowledge: it does not know that Arizona's 80% appreciation in a desert housing market differs from North Dakota's 20% in an oil economy. LLMs invert this, encoding economic knowledge from pre-training but expressing it through inference rather than exact optimization.

In Bayesian terms, OLS computes a maximum likelihood estimate in a four-dimensional linear space with no prior. An LLM holds a prior $p(\text{crash} \mid \text{state identity})$ from pre-training that encodes latent economic structure: some identifiable (industry composition, supply elasticity, subprime concentration), others unobservable and black-box. By reading the 40-state training table, the model constructs a likelihood $p(\text{features} \mid \text{crash})$ during the forward pass. No

statistics are pre-computed; the model extracts this structure from the raw data. Its prediction is a posterior:

$$p(\text{crash} \mid \text{features}, \text{state identity}, \text{context}) \propto p(\text{features} \mid \text{crash}) \cdot p(\text{crash} \mid \text{state identity})$$

We test four conditions: **Predict** (pure reasoning with real training data), **Predict + Tools** (Python with numpy/pandas, regression libraries banned), **RLM** (iterative multi-turn with Python REPL), and **Zero-data** (model receives only state names, no features or training examples). The zero-data condition isolates the prior term by removing all in-context data. A **shuffled** condition (training labels randomly reassigned across states) tests prior-likelihood conflict. Full methodology: llm_experiment.ipynb.




```
Mean RMSE - OLS:            8.88 (+/- 2.39)
Mean RMSE - LLM (real):     4.34 (+/- 0.91)
Mean RMSE - LLM (shuffled): 9.86 (+/- 3.04)
Mean RMSE - LLM (zero):     2.58 (+/- 0.72)
```

The zero-data condition changes the picture. Given only state names (no features, no training data, no role description), Claude Opus 4.6 achieves mean RMSE 2.58 (+/- 0.70), outperforming every other condition. The model has memorized state-level housing crash severity from pre-training. State-level FHFA HPI declines are among the most analyzed statistics in

modern economics, widely published in academic papers, government reports, and post-crisis analysis.

The four conditions form a hierarchy:

Condition	Mean RMSE	What it measures
Zero data	2.58	Pure recall from pre-training
Real data	4.34	Recall degraded by incomplete features
OLS	8.88	Features only, no prior
Shuffled data	9.86	Prior fought by wrong labels

Providing real training data (4 features, 40 states) degrades predictions from 2.58 to 4.34. The four features explain 58% of crash variance (see Model 2 above), meaning 42% of the information the model's prior already encodes cannot be expressed through the feature set. The likelihood term introduces a 4-dimensional bottleneck that partially overrides a stronger prior. This is a concrete instance of a Bayesian principle: when the prior is very informative and the likelihood is weak, the posterior is worse than the prior alone.

The shuffled condition (RMSE 9.86) shows the model does not ignore in-context data. Wrong labels pull it off its prior, degrading performance below OLS. The model incorporates training examples even when they are adversarial, which is why shuffling makes things worse rather than having no effect.

Method	Model	Split 0 RMSE
Predict	Claude Opus 4.6	3.16
Predict	GPT 5.4 Heavy	3.28
Predict	GPT 5.4 Pro	3.84
Predict	Gemini 3.1 Pro	3.87
Predict + Tools	GPT 5.4 Heavy	4.11
Predict	Gemini 3 Flash	4.21
RLM	GPT 5.4 Mini	4.71
OLS Regression	statsmodels	5.45

The latent space hypothesis

The leaderboard is ordered by model strength, not harness sophistication. Tools collapse the model's representational advantage. GPT 5.4 Heavy scores 3.28 without tools and 4.11 with them. With tools, it computed Washington's KNN average at -17.89. Without tools, it reasoned about the tech economy cushion and predicted -21.50, closer to the actual -25.19. Tools reduce the model's similarity function from its full latent space to the four provided columns.

Illinois provides direct evidence of latent variables at work. Every model except Opus under-predicts IL by 6+ points. The four-feature profile says "moderate Midwest state," but Opus reasons about subprime exposure on the south side, a variable absent from the dataset, and predicts within 2.81 points of the actual -21.81. GPT 5.4 Heavy misses by 6.41; Pro by 7.31.

Whether this latent knowledge comes from memorized crash statistics or from structured economic reasoning cannot be determined for this task. The 2008 housing crash is too well-documented to isolate the two mechanisms. Testing the predictive primitive hypothesis cleanly requires a target variable the model has not encountered during pre-training.

Conclusions

Pre-crisis appreciation alone explains 47% of cross-state crash variance ($R^2 = 0.47$, $p < 0.0001$). Adding fundamentals raises R^2 to 0.58 (F-test $p = 0.011$), with population growth as the strongest additional predictor ($p = 0.005$). The residuals show what four indicators miss: Michigan's auto collapse, Alaska/DC/Hawaii's insulation.

LLM experiment: memorization and prior-likelihood interaction

Claude Opus 4.6 achieves mean RMSE 4.34 across 20 splits versus OLS's 8.88. We initially interpreted this as evidence that LLMs could outperform statistical methods through latent economic knowledge. However, a zero-data condition (model receives only state names, no features or training data) achieves RMSE 2.58, revealing that the model has memorized state-level crash severity from pre-training.

This discovery reframes the experiment. The question "can an LLM outperform OLS?" is trivially answered when the model has memorized the target variable. The more interesting finding is what happens when memorized knowledge meets incomplete data. Four features explaining 58% of variance degrade the model's predictions from 2.58 to 4.34: the in-context data partially overrides a stronger prior with a weaker signal. Shuffled labels degrade further to 9.86, confirming the model incorporates training data even when adversarial. This is a Bayesian result:

$$p(\text{crash} \mid \text{features}, \text{state}, \text{context}) \propto \underbrace{p(\text{features} \mid \text{crash})}_{\substack{\text{likelihood from} \\ \text{training data}}} \cdot \underbrace{p(\text{crash} \mid \text{state})}_{\text{pre-training prior}}$$

When the prior is near-perfect (zero-data RMSE 2.58) and the likelihood is incomplete ($R^2 = 0.58$), the posterior is worse than the prior alone. This principle generalizes beyond our specific task: for any well-documented outcome, providing an LLM with incomplete features may hurt more than help.

Delivery mechanism findings

Regardless of memorization, the experiment reveals how different interfaces extract knowledge from the model. Pure reasoning outperforms tool-assisted prediction in every comparison. GPT 5.4 Heavy scores 3.28 without tools and 4.11 with them. Tools anchor the model on explicit features, collapsing its representational space from the full latent dimensionality to the four provided columns. Extended thinking also hurts: GPT 5.4 Pro's 28 minutes of deliberation (3.84) underperforms Heavy's 4 minutes (3.28). These findings transfer to settings where the model's knowledge comes from reasoning rather than recall.

Limitations

The predictive primitive hypothesis, that LLMs can function as frozen-weight prediction functions using latent structure from pre-training, remains plausible but unproven. The 2008 housing crash is too well-documented to disentangle memorization from generalized economic reasoning. A clean test requires a target variable absent from the training corpus: a novel economic indicator, a future event, or a synthetic outcome constructed from real features. This is a limitation of the current experiment, not a refutation of the hypothesis.

Reflection

To be completed.

Appendix

LLM experiment methodology and per-model analysis: llm_experiment.ipynb

Source code and data: github.com/ethanaggor/sds1730-final-project